Predicting Employee Turnover: A Statistical Analysis of Workforce Attrition Factors

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***Abstract*—Employee turnover represents one of the most persistent and costly challenges facing organizations today, yet most retention strategies rely on intuition rather than data. This paper presents a statistical analysis of workforce attrition using an HR dataset of 14,999 employee records drawn from Kaggle. We apply descriptive statistics, relational database querying (SQLite), correlation analysis, and visual analysis to identify the factors most strongly associated with employee departure. Our findings reveal an overall turnover rate of 23.81%, with low-salary employees leaving at nearly 30%—more than four times the rate of high-salary employees. Job satisfaction emerges as the single strongest predictor (r = –0.388); employees who left averaged a satisfaction score of 0.44 compared to 0.67 for those who stayed. We further identify 615 current employees who match a high-risk churn profile. These findings provide HR departments with actionable, data-driven criteria for early intervention and targeted retention strategies.**

***Index Terms*—employee turnover, workforce attrition, HR analytics, churn prediction, descriptive statistics, SQLite, data analysis**

I. INTRODUCTION

The cost of replacing an employee is widely estimated at between 50% and 200% of that employee's annual salary, accounting for recruiting, onboarding, lost productivity, and institutional knowledge transfer [2]. Yet despite this substantial burden, many organizations respond to turnover reactively—conducting exit interviews only after employees have already resigned, and implementing changes too late to retain the individuals who prompted them.

The central challenge is one of prediction: given observable workplace factors, can an organization identify which employees are most likely to leave before they submit their resignation? Modern HR systems routinely capture variables such as satisfaction surveys, performance evaluations, compensation tiers, workload metrics, and promotion histories—precisely the kinds of signals that may indicate attrition risk.

This paper addresses that challenge using a dataset of 14,999 employee records. We pursue four research questions: (1) How strongly does salary level predict turnover? (2) What is the relationship between satisfaction and departure? (3) Does workload influence turnover? (4) Can we identify high-risk employees using observable characteristics?

Our approach is grounded in accessible statistical methods—descriptive statistics, group comparisons, correlation analysis, and SQL querying against a normalized relational database—rather than black-box machine learning models, making findings interpretable and actionable for HR practitioners.

II. RELATED WORK

Early work by Mobley [1] established a foundational behavioral model of turnover, identifying job dissatisfaction as the primary precursor to resignation. Griffeth, Hom, and Gaertner [2] confirmed through meta-analysis that satisfaction, pay, and workload are the most reliable predictors of departure across industries and time periods.

More recent computational approaches have applied supervised machine learning to HR datasets. Punnoose and Ajit [5] applied random forests and logistic regression to the same Kaggle HR dataset, achieving accuracies above 97%. Alduayj and Rajpoot [7] compared decision trees, SVMs, and neural networks, finding tree-based models consistently outperformed others on HR tabular data. Zhao et al. [6] demonstrated ensemble methods improve performance but at significant cost to interpretability [8].

Our work prioritizes interpretability over predictive accuracy: every finding can be expressed as a SQL query, a summary statistic, or a chart that any analyst can reproduce and explain to a non-technical stakeholder.

III. DATASET

A. Source and Scope

The dataset is the HR Employee Retention dataset from Kaggle [9], containing 14,999 employee records. The outcome variable (*left*) indicates whether the employee departed (1) or remained (0).

B. Variables

The dataset includes ten variables across four domains (Table I).

TABLE I. DATASET VARIABLES

Domain	Variable	Type	Descr.
Satisfaction	satisfaction_level	Num [0,1]	Job satis faction
Performance	last_evaluation	Num [0,1]	Perf. review
Workload	number_project	Integer	Projects
Workload	avg_monthly_hrs	Integer	Avg. hrs/mo
Tenure	time_spend_company	Integer	Yrs at company
Safety	work_accident	Boolean	Work accident
Career	promotion_last_5yrs	Boolean	Promote d 5 yrs
Compensation	salary	Category	Lo/med/ hi
Outcome	left	Boolean	Left company

C. Data Quality

A missing value audit revealed no null entries. Outlier detection using the IQR method identified anomalous values in avg_monthly_hours (above 310/month), retained as plausible extreme-workload cases. Schema CHECK constraints validated satisfaction and evaluation scores within [0,1], non-negative hours and tenure, and foreign key consistency across all four tables.

D. Database Schema

The flat CSV was normalized into a four-table SQLite schema [10]: (1) *salary_levels*—lookup table; (2) *employees*—core identity table; (3) *performance_records*—satisfaction, evaluation, hours; (4) *employment_events*—accidents, promotions, turnover outcome. This eliminates redundancy, enforces referential integrity, and allows the turnover outcome to be joined against any attribute combination in a single query.

IV. METHODOLOGY

A. Descriptive Statistics

For all numeric variables we computed mean, median, standard deviation, variance, range, Q1, Q3, and IQR, calculated separately for employees who left and those who stayed.

B. Group Comparison

Turnover rates were computed for each level of categorical variables (salary tier, promotion status, work accident). Numeric variables were segmented into bins to identify non-linear patterns.

C. Correlation Analysis

Pearson correlation coefficients were computed between each numeric feature and the binary *left* outcome, producing a ranked list of predictors by association strength.

D. SQL-Based Analysis

All group-level queries were executed against the normalized SQLite database: overall turnover rate, turnover by salary tier (three-table join), average satisfaction and monthly hours by turnover status, and high-risk identification using compound WHERE filters.

E. Risk Profiling

A high-risk profile was defined using three simultaneous criteria: satisfaction below 0.4, salary tier 'low', and no promotion in the last five years, constrained to employees still at the company.

V. RESULTS

A. Overall Turnover Rate

Of 14,999 employees, 3,571 left, yielding an overall turnover rate of **23.81%**—substantially higher than the U.S. industry average of approximately 15% [4].

B. Turnover by Salary Tier

Table II shows low-salary employees leave at over four times the rate of high-salary employees.

TABLE II. TURNOVER RATE BY SALARY TIER

Salary Tier	Total	Left	Rate
Low	7,316	2,172	29.69%
Medium	6,446	1,317	20.43%
High	1,237	82	6.63%

C. Job Satisfaction and Turnover

Job satisfaction shows the most pronounced difference between retained and departed employees (Table III) and produces the strongest Pearson correlation with turnover (r = –0.388). Satisfaction differs by 0.227 points—a 34% relative gap. By contrast, evaluation scores are nearly identical (0.715 vs. 0.718), confirming turnover is a satisfaction problem, not a performance problem.

TABLE III. AVERAGE FEATURE VALUES BY TURNOVER STATUS

Feature	Stayed	Left
Satisfaction level	0.667	0.440
Last evaluation	0.715	0.718
Number of projects	3.79	3.86
Avg. monthly hours	199.1	207.4
Years at company	3.38	3.88

D. Feature Correlation Analysis

Table IV ranks each feature by its Pearson correlation with the *left* outcome.

TABLE IV. FEATURE CORRELATION WITH TURNOVER

Feature	Pearson r
satisfaction_level	–0.388
work_accident	–0.155
time_spend_company	+0.145
avg_monthly_hours	+0.071
promotion_last_5years	–0.062
number_project	+0.024
last_evaluation	+0.007

E. Workload and Turnover

Employees who left worked an average of 207.4 hours/month compared to 199.1 for those who stayed—8.3 extra hours, or roughly two additional workdays per month. The correlation of avg_monthly_hours with turnover (r = +0.071) suggests overwork is a contributing factor, acting as a compounding stressor alongside low satisfaction and low pay.

F. High-Risk Employee Profiling

Applying the compound risk profile identified **615 current employees** matching all three criteria (Table V).

TABLE V. SAMPLE HIGH-RISK EMPLOYEES (TOP 10)

Emp. ID	Satisfaction	Hrs/Mo	Tenure
2087	0.12	244	5 yrs
3129	0.12	257	6 yrs
4026	0.12	241	2 yrs
4684	0.12	276	4 yrs
5336	0.12	166	3 yrs
7772	0.12	161	4 yrs
8745	0.12	287	4 yrs
9283	0.12	200	3 yrs
11069	0.12	191	5 yrs
13280	0.12	191	5 yrs

VI. DISCUSSION

A. Compensation is the Dominant Structural Factor

The four-to-one ratio in turnover rates between low and high salary tiers is the single most actionable finding. Compensation structure creates a systemic attrition risk that cannot be resolved through engagement programs or managerial coaching alone. Organizations with large proportions of low-salary employees should treat compensation review as a primary retention strategy [2].

B. Satisfaction Predicts Turnover Independently of Performance

The near-identical evaluation scores between leavers and stayers (0.718 vs. 0.715), combined with the near-zero correlation of last_evaluation (r = +0.007), challenges the common assumption that turnover is driven by underperformers. This replicates Mobley's foundational finding [1]: retention efforts should be framed as a satisfaction and engagement problem, not a performance management problem.

C. Workload Acts as a Compounding Risk Factor

The highest-risk employees combine low satisfaction with above-average hours, suggesting overwork amplifies the effect of dissatisfaction rather than driving attrition independently. HR interventions should target the intersection of these factors [3].

D. Early Intervention Opportunity

The identification of 615 high-risk current employees is the most practically valuable output of this analysis. HR departments can operationalize it as a recurring SQL query to generate a prioritized list for manager outreach, salary review, or promotion consideration before employees resign.

E. Limitations

This study has several limitations. First, the dataset is simulated [9] and limits generalizability of specific threshold values. Second, the analysis is descriptive, not causal. Third, the dataset lacks department or role information. Fourth, class imbalance (~24% leavers) must be addressed before extending this framework to predictive modeling.

VII. CONCLUSION

This paper analyzed an HR dataset of 14,999 employees to identify factors most strongly associated with workforce attrition. Salary tier and job satisfaction are the two dominant predictors—satisfaction producing the strongest correlation ($r = -0.388$)—while performance evaluation is essentially uncorrelated with departure ($r = +0.007$). Critically, departed employees were equally well-evaluated as retained employees, confirming that turnover is a satisfaction problem, not a performance problem.

By normalizing a flat HR CSV into a constraint-validated SQLite schema and operationalizing findings as reusable SQL queries, we demonstrate that organizations can generate actionable, prioritized intervention lists from existing HR data without requiring machine learning infrastructure. The 615 high-risk employees identified here represent an immediately addressable retention opportunity.

Future work should apply this framework to real organizational data with department-level granularity, incorporate time-series analysis, and explore causal inference methods to move beyond association toward actionable causal claims about salary increases or promotion effects on retention.

ACKNOWLEDGMENT

The author thanks the faculty of the Department of Computer and Electrical Engineering and Computer Science at California State University, Bakersfield for guidance on this project.

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